Discover Artificial Intelligence

Review

A review of large language models and the recommendation task

Jacob Munson[1] · **Thomas Cuezze**[1] · **Siddat Nesar**[2] · **Dominique Zosso**[3]

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Abstract
Recommender systems are now ubiquitous across the internet, from streaming services to online shopping to social media. Traditional systems operate behind the scenes, often invisible to the end user. While these systems have enjoyed prolific success, they have limitations—namely, their mechanical interactions lack contextual awareness. A promising area of research is the combination of large language models (LLMs) with traditional recommendation methods to increase flexibility and performance. We discuss prominent examples, including conversational recommender systems, LLMs as end-to-end recommenders, and LLMs as encoders for recommendation. Of particular importance is the transformer neural network architecture, which underpins these LLMs and has shown itself to be incredibly powerful in natural language processing, and has now been adapted to serve recommendation tasks. This review offers a unique perspective on the evolving role of data in recommender systems, tracing data requirements from traditional matrix-based data and knowledge-based data, to the adoption of transformers with web-scale data. We detail how these shifting data paradigms have shaped the field and integration of transformer architecture, large language models, and chatbots in modern recommender systems. This paper is intended for readers interested in the intersection of recommender systems and transformers (and LLMs), those tracking the evolution of data used in such systems, and newcomers seeking an introduction to these topics.

Keywords Transformers · Large language models · Prompting · Chatbots · ChatGPT · Bard · Claude

1 Introduction

Recommender systems (RS) are incredibly common in day-to-day life. From video and music streaming to e-commerce and social media, these systems are felt far and wide in nearly everyone's online presence [1]. Online platforms collect user interaction data, such as clicks, views, listens, or purchases, and use this data to make personalized recommendations. These are used in menus, push notifications, email reminders, and more. Using interaction data, the systems match users (e.g., viewers and customers) and items (e.g., videos, goods, etc.). Recommender systems drive user satisfaction, manage information overload, and enhance user engagement [2, 3]. Collecting data over time helps improve future recommendations, as user preferences and item attributes are better understood.

Jacob Munson and Thomas Cuezze have contributed equally to this work.

✉ Jacob Munson, jacobmunson1@montana.edu; Thomas Cuezze, thomas.cuezze@student.montana.edu; Siddat Nesar, siddatnesar@montana.edu; Dominique Zosso, zosso.dominique@gmail.com | [1]Department of Mathematical Sciences, Montana State University, P.O. Box 172400, Bozeman, MT 59717, USA. [2]Electrical and Computer Engineering, Montana State University, P.O. Box 172400, Bozeman, MT 59717, USA. [3]Zoot Enterprises Inc., 555 Zoot Enterprises Ln, Bozeman, MT 59718, USA.

Traditionally, the data collected from RS, such as clicks, views, listens, or purchases, can be either *explicit* or *implicit* feedback [1, 4]. Explicit feedback, such as ratings or likes, is submitted by the user as an *explicit* indication of preference [5]. Implicit feedback, such as views or listens, indicates item consumption feedback that *might imply* user preference [5]. For example, repeated consumption of a film, show, or song likely indicates a user enjoys that item, but the user has not explicitly rated/liked the item. In this traditional characterization of RS, the mode of interaction is the user interacting with the system through their clicks, purchases, views, and so on.

While RS have been wildly successful in using historical user data to produce recommendations, the range of acceptable interactions remains relatively limited, with many popular approaches notably ignoring complex feedback such as text [6]. Addressing this limitation, conversational recommender systems have emerged, shifting from a reliance on historical user feedback to human-system dialogue exchanges, where the system often has access to external relevant knowledge, such as webpages or relevant databases [6]. Exemplifying this shift, this review focuses on the application of the transformer architectures, especially Large Language Models (LLMs), to recommender systems. These models are naturally suited for the recommendation task as they are capable of accepting a diverse range of inputs (e.g., historical user data, descriptions of a user's interests, descriptions of items, etc.) and producing a diverse range of outputs (e.g., direct recommendations, rating predictions, lists, etc.).

We review classical RS techniques in Sec. 2.1-2.4, discuss knowledge-based RS in Sec. 2.5, provide a broad overview of conversational RS in Sec. 2.6-2.8, explore recent language modeling advances in Sec. 3 in preparation for describing the intersection of language models and recommendation in Sec. 4, and finally we offer conclusions in Sec. 5. The abbreviations used are given in Table 1.

2 Recommender systems

Classical recommender systems are often broken into two main categories: content-based methods and collaborative filtering. The two approaches use different data types and modeling frameworks, and can be used complementarily, forming so-called "hybrid" methods. We briefly review these classical frameworks before extending our discussion into more esoteric areas of RS, including knowledge-based recommender systems, which primarily focus on incorporating external data into the system, and conversational recommender systems, focusing on the conversational mode of interaction. This setup ultimately serves to frame a broader discussion of exploiting large external data for recommendation, which naturally leads into using LLMs for recommendation. While we provide a broad overview of RS, we encourage readers to visit [7] for a more in-depth review of modeling trends.

Table 1 Common Abbreviations in This Paper

Abbr.	Meaning
RS	Recommender System
LLM	Large Language Model
CB	Content-based (methods)
CF	Collaborative Filtering
MF	Matrix Factorization
DL	Deep Learning
RMSE	Root mean-squared error
MAE	Mean absolute error
KB	Knowledge-based
LOD	Linked Open Data
CRS	Conversational RS
NLP	Natural Language Processing
LM	Language Model
PLM	Pre-trained Language Model
SOTA	state-of-the-art
NDCG	Normalized Discounted Cumulative Gain

2.1 Content-based methods

Any recommendation method that takes into account the actual content of the items being recommended is a content-based method. Content-based (CB) methods rely on a "feature-based representation of the content of recommendable items" to make recommendations [8]. Item content (e.g., film genre, director, etc., in the case of film) is used as input into CB methods, as illustrated in Fig. 1 (Content-based). While CB methods are historically less used than collaborative filtering methods, conversational recommender systems (Sec. 2.6) rely heavily on item content and descriptions, as they often operate with little historical user data [9, 10]. For a broad view into CB methods through time, see [8, 11, 12].

2.2 Collaborative filtering

In contrast to content-based methods, collaborative filtering approaches (CF) rely on user feedback (e.g., ratings) collected from many users, used *collaboratively*, "without need for exogenous information about either items or users" [4]. Feedback is usually organized into a user-by-item matrix, called the *ratings matrix*, where recorded feedback (e.g., a rating) associated with a particular user-item pair is placed into the matrix at the user row and item column and missing values (i.e., feedback that has not been observed) is left blank [1]. As only a small portion of possible user-item combinations are observed, since most users have not consumed most items, the ratings matrix is typically very sparse or *incomplete*. Figure 1 (Collaborative filtering) illustrates this workflow.

Given this common framing of the feedback data, matrix factorization (MF) techniques have been widely applied in CF. The ratings matrix is decomposed into dense latent factor matrices, which can be recombined to produce a dense (prediction) ratings matrix. Predictions can then be made per user and items that the user is predicted to rate highly can be displayed to the user: this is the TopN/ranking task of suggesting items to a user [13].

While MF techniques have been wildly successful in CF, neural network approaches have risen into prominence [7, 14, 15] given the flexibility of modern deep learning (DL) methods, often being able to incorporate side information or jointly learning CB and CF predictors [16–18]. These modern approaches are often hybrid in nature, allow the use of multiple data sources and types.

2.3 Hybrid methods

Diverse RS models, such as CB and CF approaches, can be used synergistically in *hybrid methods*. For example, CF suffers from the *cold start* problem, where users/items with few interactions (perhaps zero, for new users/items) are difficult to

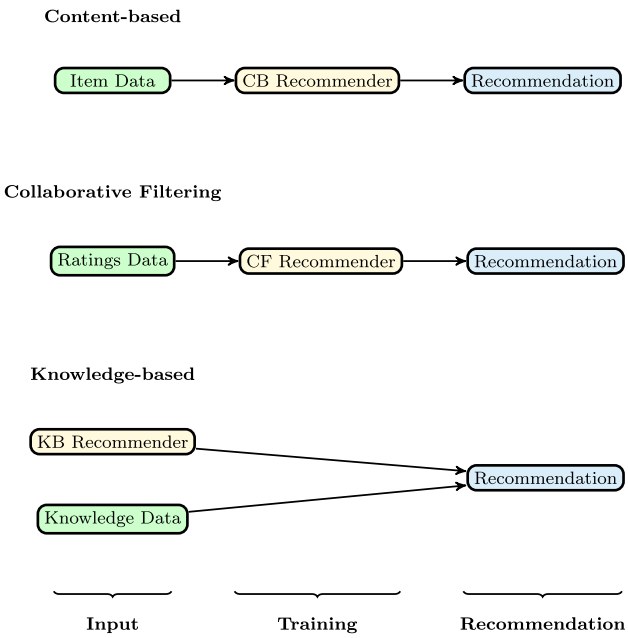


Fig. 1 Three common recommendation approaches. Content-based methods use item data to make recommendations. Collaborative Filtering methods use ratings data to make recommendations. Knowledge-based methods typically do not require training data, as they are often rule- or logic-based, but the knowledge data is used as reference data to make recommendations

generate recommendations for. However, CB methods use item information to produce recommendations and thus do not suffer from the cold start problem. In this way, CB methods can "help" a CF system when users/items are new.

Combining constituent models into a hybrid RS can be done in several ways as outlined in [19]. First, the predictions offered by the constituent models can be simultaneously presented in a *mixed* fashion, where multiple sets of top predictions are given at once. Second, and similarly, predictions from separate RS can be *weighted* (e.g., a linear combination of predictions), so that top predictions from different models are not equally weighted. Third, hybrid RS can *switch* between predictors, selectively using the appropriate constituent model (e.g., CB for new users/items and CF otherwise). Fourth, *feature combination* approaches combine diverse data (e.g., ratings and item description data) into one dataset and use one RS capable of accepting the complicated data. Fifth, hybrid RS can *cascade* the output of one RS as the input for another RS; this "refines" recommendations at each cascade. Sixth, similar to cascade, output from one RS can be used as supplemental input in a *feature augmentation* capacity into another downstream RS. Lastly, again similar to cascade, a *meta-level* RS takes output from one RS as the primary, but not sole, input for a subsequent RS. The cascade, meta-level, and feature augmentation approaches differ primarily in how input data (from a previous RS) is used: for cascade/meta-level/feature augmentation, previous RS output is the sole/primary/supplemental input to the next RS.

2.4 Data splitting and evaluation

Effective use of RS is highly dependent on both the data splitting and evaluation schemes [20]. Data splitting, the process of separating the data into train and test sets, is an important step in producing a useful RS. Evaluation, which depends heavily on how the data is split, is important to determine both the effectiveness of the RS and the expected performance going forward.

2.4.1 Train and test sets

RS model training includes the crucial step of producing train and test sets [21]. The two main data splitting decisions are (1) the relative amount of data in train/test sets and (2) any temporal ordering of the split. A common split is the standard "80/20" train/test split regime,[1] ignoring temporal ordering. Despite its common use, misleading results are likely.

For example, consider a standard 80/20 split in which some test observations are inadvertently "easy" to predict. Consider a user with training data for the film *The Mummy Returns* (2001) and test data for *The Mummy* (1999). It is evident that a user who highly rates *The Mummy Returns* almost guarantees the user would rate *The Mummy* highly. The underlying problem is that *The Mummy*, in this context, is being unfairly evaluated as the RS is predicting the past given the future.

Respecting temporal order might involve producing a train/test split around a specific date; data before (after) the date is used for training (testing) [21]. However, RS has the additional complexity of being centered around users. Consequently, user-centered evaluation might be considered, allowing train/tests sets to be built around users (e.g., "N ratings per user are used for testing"). A user-centered evaluation can be used in conjunction with a temporal dependence (e.g., "the *last* N ratings per user are used for testing"). This ensures that earlier (later) data is put into the train (test) set per user without the need to establish a global train/test split date.

2.4.2 Evaluation

Evaluation is a diverse and complicated area of study in its own right and depends highly on the application and data split scheme. To underscore the diversity of end goals, consider the following RS conundrums: (1) an e-commerce RS: should the focus be increasing sales, increasing time on platform, or obtaining clickthroughs on email offers, (2) a streaming RS: should the focus be maintaining high engagement, increasing time on platform, or ensuring users are satisfied with recommendations (e.g., users submit explicit feedback on recommendations), or (3) a social media RS: should the focus be maintaining high engagement or increasing time on platform even if a user is primarily engaged in polarizing content?

These scenarios highlight the complexity and variability of RS evaluation, emphasizing the need for a clear understanding of the goals of the system. Many systems, such as those based on MF, rely on a minimization objective between a prediction and a training value, thereby minimizing errors. It is natural, though problematic, to evaluate RS via error measures, such as root mean-squared error (RMSE) or mean absolute error (MAE) on the train and test sets [20].

[1] This indicates 80% (20%) of the data is used for training (testing). Variations, such as 75/25 or 90/10, are also common.

While the process of producing accurate (low RMSE/MAE) predictions is important, producing predictions for train/test set observations does not help surface new content to users: the Top-N task is responsible for surfacing recommendations of unseen items ("lists") to users. To produce recommendations on unseen items for users, predictions are made for each user across a large set of prospective items unseen to the user in either the train or test sets. Predictions can then be sorted per user and items with higher predictions are given priority for presentation to the user; it is reasonably argued that highly rated items should be more relevant to the user.

Top-N performance is often measured by a system's ability to produce relevant recommendations, where a user's test set data is considered relevant [22]. Recovering test set data in a Top-N list suggests a model is capturing a user's true preferences. However, RS training is doubly complicated as low train/test errors (i.e., accurate predictions) do not imply strong Top-N performance [23–25]. A recent survey on RS evaluation can be found in [26]. The model paradigms discussed thus far generally optimize a single objective—typically minimizing the squared loss, as is common in CF matrix factorizations or CB regression methods—but it is worth the existence of multi-objective RS [27]. While these systems are beyond the scope of this work, there exist approaches that aim to balance multiple objectives, such as prediction accuracy and Top-N measures, such as diversity.

2.5 Knowledge-based recommender systems

While CF uses ratings matrix data generated by users and CB uses item features devoid of context about user interactions, both CF and CB require data during training. Knowledge-based (KB) systems can be thought of as a third, orthogonal RS type using *external* reference data exploited through logic and rules. Figure 1 (Knowledge-based) illustrates this workflow. Early KB models offered domain flexibility and deep product base knowledge [28]. Despite these advantages and the well-documented shortcomings of CF [29, 30], such as the "cold start" problem [31] of new users having no data, KB systems received little research attention in comparison to CF methods. We briefly discuss KB systems and the evolution of RS using external data, setting the stage for both Conversational RS using various external data sources and LLM-based RS that use truly massive, web-scale data. KB approaches leverage reference knowledge, such as catalogs or menus, often in *Knowledge Graph* form [32], that can be exploited through rules or filtering strategies, where a user might incrementally communicate their preference to a system, often conversationally [33].

2.5.1 Knowledge graphs

Early KB systems focused on well-defined domains, such as menu choices, films, or even apartment-finding [34]. The self-contained nature of these domains allowed for explicit relationships defined in knowledge graphs to be useful; these explicit relationships can aid interpretability [35–39]. While KB systems are flexible, their reliance on well-defined knowledge graphs is inherently limiting. For example, an e-commerce store might use a KB system for electronic goods, but it will be unable to offer comparisons/descriptions of products beyond its typical inventory, even for products highly similar to theirs.

While knowledge graphs usually rely on expensive manually engineered features, recent advancements in machine learning, including DL approaches, have used knowledge graph embeddings [40–43]. These embeddings are useful in a variety of machine learning tasks, such as link prediction [44–46], organizing and recommendation of Wikipedia articles [47], and explainability [36, 48, 49]. Moreover, even though most knowledge graphs are typically correct, their completeness is often not guaranteed as the manual engineering is often unreliable [50–53].

2.5.2 Linked data

Predefined knowledge graphs inherently limit the ability of KB systems to provide recommendations outside a narrow focus, thus it is natural to explore and connect to graphs across providers: the concept of *linked data* arises to create a network of interconnected information by combining various knowledge graphs [54].[2] Linking knowledge graphs helps expand the scope of information available while producing a more comprehensive and interconnected web of knowledge exploitable by machine learning models, such as RS [61]. By linking knowledge graphs, the information about particular

[2] Knowledge graphs are often constructed from structured knowledge bases, such as DBpedia [55], Never-Ending Language Learning (NELL) [56], Wikidata [57], Yet Another Great Ontology (Yago) [58], Knowledge Vault [59], and the (now defunct) Microsoft Academic Service (MAS) [60].

items is extended (e.g., better descriptions of the same item in two different knowledge graphs) and cross-domain recommendations (e.g., a user prefers certain books, but is currently seeking film recommendations) become possible.

While linked data is invaluable, collecting and organizing such data is nontrivial. The Linked Open Data (LOD) project provides a framework for standardizing and publishing data [62, 63]. The standardization allows for ease of use, while open access allows for free use and linking with other datasets. This openness facilitates a culture of data sharing, breaks down data silos, and helps to produce a global network of interconnected datasets. This interconnectedness has proven particularly beneficial for RS, where semantic understanding is crucial [19]. This represents a shift from a web built for humans to a web built for humans and machines.

A system's ability to interpret the meaning and relationships embedded within data, or *semantic understanding*, helps to more accurately portray user preferences and item characteristics. The vastness of LOD enables RS to go beyond surface-level recommendation, as systems gain deeper understanding of user preferences and item characteristics, contributing to more nuanced and accurate recommendations. As knowledge graphs have demonstrated their utility for RS, LOD has similarly become an invaluable resource, enabling more nuanced semantics-aware recommendation. Moreover, LOD helps improve item descriptions through *direct item linking* and *item description linking*, enriching the content and enhancing the overall quality of recommendations [19].

2.5.3 LOD-enabled recommender systems

Deeper relationships are often lost in data without semantic understanding. For example, in a film recommender with works from directors *A* and *B*, which have many non-superficial (but not textually represented) similarities, a user's preference for director *A* reasonably indicates the user might enjoy director *B*, but the lack of explicit overlap is a struggle for standard CB methods [19]. The integration of rich, LOD-based semantic data into CB, KB, and hybrid RS models is an important development in advancing semantics-aware RS [12]. While semantics-aware RS predates the emergence of LOD, the semantic power of LOD has enabled meaningful advances [64–70]. Systems built on LOD serve users across the gamut of applications by leveraging cross-domain functionality [31, 71], incorporating side information [72, 73], handling implicit feedback [74], tackling issues like the cold start problem [31, 75], making low stock retail (secondhand books) recommendations [76], aiding feature selection [77], and predicting graph links [78], as well as general improvements [79–85]. See [19] for a quality discussion of the intersection of RS and LOD.

2.6 Conversational recommender systems

Thus far, the choice of recommender model can be perfectly delineated by the data used: collaborative filtering uses ratings matrix data, content-based methods use item features, knowledge-based models use knowledge data, and hybrid models combine data sources. This backdrop of approaches, along with their diverse data requirements, allows for Conversational Recommender Systems (CRS) to emerge as a unifying recommendation framework. Figure 2 is provided to illustrate a broad characterization of the evolution of RS modeling approaches and data availability.

Different from previous modeling frameworks, CRS is not defined by the data type, but rather the conversational mode of interaction [6, 86, 87]. Recommendations are given by conversational agents existing across a continuum of interactivity, from rigid and mechanical, featuring menu options, clickables, and limited opportunities for creative user input and rote outputs, to free-form natural language conversational agents with access to diverse public knowledge [6, 86]. This spectrum of interaction styles, both from user and system, makes CRS a diverse and challenging field with enormous potential impact [6, 86, 87].

In this work, we ultimately explore the RS and LLM intersection and thus focus on CRS models that utilize natural language, as opposed mechanistic, rule-based approaches. Even within this more narrow characterization of CRS, the style of interaction can vary considerably; some systems allow for fluid conversation and others operate in a very strict "question and answer" form. Regardless, conversations serve as the process through which users describe their interests and agents follow up with information and questions in a multi-turn fashion, and ultimately recommendations are made; an agent is expected to honor the user's disclosed (dis)interests while exploiting its domain knowledge and/or training data [88, 89]. This dynamic operates in contrast to traditional RS in which user preferences are wholly expressed (via existing feedback or historical behaviors) prior to the current session. In CRS, users express their preferences during the current session.

These complex systems feature two components: a frontend and a backend. The frontend is an input/output interface and the backend processes input and produces recommendations.[3] A CRS usually appears as one unit to the user [86]. The distinction, however, presents important technical considerations. For example, no matter how advanced a backend, a frontend that is limited to pre-selected responses (e.g., "Yes", "No", "Something else", etc.) inherently limits user expressibility. Conversely, an over-accommodating frontend is useless without a capable backend. A flexible backend/frontend combination accepts diverse forms of user input dialogue (e.g., direct inquires and "chit-chat"), interacts with user input to ensure enough information is available, produces quality recommendations by appealing to some combination of user input, user profiles, and domain knowledge, and finally surfaces those recommendations in an appropriate manner (e.g., "You might like X, because you shared Y preference detail.") [86].

Given the complexity and potential reach of these systems, academic workshops at premier conferences, including RecSys [90–94] and CIKM [95], have aimed to consolidate knowledge, unify the field, and chart future development.[4] Early CRS developments were closely related to KB approaches, which could more easily be structured in a conversational manner [34]. However, recent natural language processing (NLP) advances have pushed the CRS landscape towards LLM-focused approaches [6]. It has been noted that producing a capable CRS might *seem* more tractable than producing a general conversational artificial intelligence (AI) system. However, interacting with a natural language conversational agent for recommendation introduces many challenges, such as handling ambiguous user inquiries, understanding intent, ensuring natural language understanding, and producing explanations [6]. Explanations, in particular, are difficult to effectively use in RS [96–98], but explanations in CRS are particularly beneficial, where providing users with a "good" reason for the recommendation tends to increase their opinion of even a poor recommendation [99].

2.7 Virtual assistants and recommendation

Perhaps the most common way in which a typical user interacts with CRS technology today is through virtual assistants such as Apple's Siri, Amazon's Alexa or Google's Assistant. Providing recommendations is a common task asked of these systems (e.g., "Hey Siri, where should I go in town for a romantic dinner tonight?"). However, a gap exists between the needs of virtual assistants and the current state of RS [100]. Virtual assistants typically interact with users through voice conversation. Users therefore expect to interact with these systems as they would with a human. In the context of recommendation, a user might ask about the system's reasoning and bounce ideas off it, with the system expected to ask questions back. User preferences may evolve over the course of the conversation. The built-in recommendation abilities of virtual assistants are not well suited to this task. They are typically incapable of providing explanations or querying the user. Moreover, once recommendations are made the systems are often incapable of refining the recommendation, leaving the user the choice of accepting the recommendation or beginning a new query.

LLMs and CRS can provide solutions to many of these issues. LLM-based RS have demonstrated remarkable ability to explain the reasoning behind their recommendations [101]. Since LLMs are capable of asking questions and incorporating responses, their recommendation process resembles natural dialogue. They are also able to model the user journeys through various interests over long periods of time [102], providing a more personal experience. While issues still exist, such as ensuring recommendations are fair [103] and preventing the model from providing false or misleading information (colloquially known as "hallucinating"), LLM-based virtual assistants show great promise. Recently, Meta Labs held a tutorial session at the 2023 Knowledge Discovery and Data Mining (KDD) conference on LLM and digital assistant integration [104] and Google announced Bard chatbot integration (based on the Gemini LLM) into their digital assistant [105].

2.8 Evaluating conversational recommender systems

CRS evaluation remains a challenge. Classical RS evaluation has algorithmic origins [20], with metrics such as RMSE being used to measure prediction accuracy. Modern approaches have adopted more user-centered techniques, often adopting statistical tools derived from psychology and sociology to measure user opinions about recommender systems [86, 106]. However, the application of these systems to CRS is understudied. For one, what should an evaluation of a CRS even

[3] CRS backends can be composed of any RS, such as CF, CB, or KB methods, or other ML models. The backend performs the recommendation; the frontend is only an interface for the user.

[4] Some workshops, such as the third [91] and fourth [93] are available online, at https://www.youtube.com/watch?v=kswpGDXuPZQ and https://www.youtube.com/watch?v=kCZ5pRMwUgo, respectively.

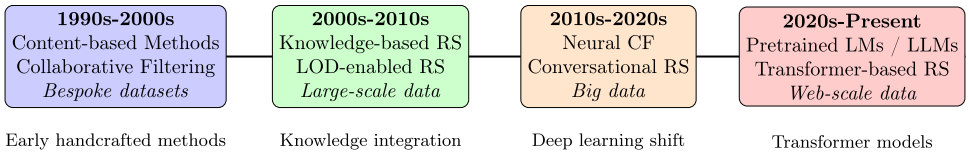


Fig. 2 Broad characterization of the evolution of modeling frameworks and *data type and scale*. Each block offers a fuzzy boundary of when a particular when two (for sake of example) modeling approaches (and the associated *style of data*) came into prominence. Naturally, the use of such models and datasets extends in the following time frame as well (e.g., while CF originated and became popular in the 1990 s-2000 s, CF is extremely popular to this day). We see as time has gone on, the style of data has shifted considerable, with models using small, bespoke datasets, to current pretrained language models using web-scale data (later discussed in Sec. 3)

seek to measure? Recommendation accuracy and efficiency, conversational aptitude, and user attitudes are all possible metrics – there is no consensus on how to balance these.

Some CRS evaluation approaches split the task into two component parts: conversation and recommendation [103, 107, 108]. This reflects the origin of CRS as the intertwining of recommender systems with NLP and allows for the use of well-developed metrics from both sides. From NLP, the BLEU (BiLingual Evaluation Understudy) score [109] and various derivatives are widely used [110]. This score was developed in the context of machine translation and is based on comparing output to a set of ground truth references. In the context of CRS, it can be used for evaluating both the overall quality of the system's conversational output and its own explanation of its recommendations more specifically. On the recommendation side, standard metrics include RMSE and MAE for rating prediction and precision and recall for item recommendation, among others [20].

The combination of these metrics from the worlds of NLP and RS has several advantages. It can be done entirely offline, avoiding the difficult, time-consuming and expensive process of conducting trials with real human users; it provides objective and quantifiable scoring, allowing for easy comparison of models and is useful in model development; and it allows for the evaluation of each of the component model parts independently. However, this approach also comes with major limitations [87] due to misalignment between what these scores are measuring and what users desire. These mathematical metrics are often based on simplistic assumptions and fail to accurately predict user sentiment. Because the nature of recommendation in the context of a conversation with a human user is so different from the simple acts of rating prediction or item ranking, it's difficult to quantify a model's performance. Output that is useful to one user may be disliked by another. A longer conversation could indicate the model is struggling to provide a good recommendation, or it could mean the user is very engaged and wants to keep going [6].

Given the limitations of these metrics, recent advances have moved towards incorporating users into evaluation. One challenge is the lack of data on how humans make recommendations to each other. Some works have focused on collecting large datasets of human conversation on which to train and evaluate models [111–113]. Other works have focused on updating evaluation methodology. The ResQue framework, first proposed in 2011 [106], has been widely adopted for RS evaluation. This very general methodology relies on a psychometric questionnaire. Recently, the CRS-Que framework has been proposed to extend this system to CRS. These systems focus on questioning users about their opinions on the system rather than using algorithmic metrics. While much more expensive, this has obvious advantages for real-world implementation.

Another area of interest is user simulation [114]. In this approach, user profiles are generated and then dialogue based on these synthetic profiles is fed into a CRS. While this does not solve the overall problem of what to measure, it does allow for the easy evaluation of the CRS in a multi-turn, interactive context. The newly proposed iEvaLM framework [115] uses LLMs as both user simulators and scorers, training one LLM to interact with the CRS as a user while another is asked to score the conversation in comparison with a reference. Overall, the evaluation of modern, LLM-based CRS is a rapidly changing field where consensus is still elusive.

3 Recent advances in language modeling

The broader NLP community has achieved considerable advances in recent years with remarkable success across a wide range of tasks [116, 117]. From unprecedented natural language understanding [117], to knowledge-base interactions [118, 119], to "fill in the blank" answers [120, 121] on diverse sets of questions, recent advances are staggering. These

advances center around the *transformer* neural network architecture [122]. The *attention* mechanism, introduced in [123] and made popular in [122], is the linchpin of the transformer architecture visualized in Fig. 3.[5]

Transformers come in three flavors: encoder-decoder, using both sides of the architecture, encoder-only, using the left side, and decoder-only, using the right side. Encoder-decoder networks (the full architecture in Fig. 3) take an input sequence, encode and decode the sequence, and produce an output sequence [124]. These networks are useful for tasks such as machine translation, where text is translated from one language to another [122]. Encoder-only networks (Fig. 3 — left) take in natural language and produce encodings for use in subsequent tasks; these networks are said to learn language *context* [125]. Decoder-only networks (Fig. 3 — right) take in and output natural language; these are common in modern chatbots, where users engage in dialogue [116, 117, 126].

The transformer has shown itself to be incredibly versatile. Scaling parameters [127–129] to create so-called *Large Language Models* (LLMs) has been particularly effective, though relatively small models have proven capable on many tasks [130–132]. We dedicate this section to discussing transformer-based models and advances driven by the transformer, setting the stage for the application of these models to the recommendation task in Sec. 4. Transformer-based LLMs are rapidly evolving, so we provide only a glimpse into the environment. See [133, 134] for recent surveys.

3.1 (Not so) large language models

While the transformer architecture underpins large language models, earlier transformer models, especially BERT (Bidirectional Encoder Representations from Transformers) [125] in late 2018, were tiny[6] by comparison. Though small, BERT's influence on the NLP community is hard to overstate. Foremost, BERT helped drive the rise of *pre-trained language models*[7] [136] and more generally, *foundation models* [137], which are powerful, versatile tools pre-trained on massive data. For instance, BERT is pre-trained on the BookCorpus (800M words) [138] and English Wikipedia (2.5T words) [139] datasets with two pre-training objectives: (1) Masked Language Modeling, in which the model predicts a masked (i.e. held out) token from a sentence (considering the left and right sides of the token) and (2) Next Sentence Prediction, in which the model predicts if one sentence follows from another.

This dynamic typifies the trend in which models are "taught" (via pre-training) a base language on a large text corpora, thereby creating a general NLP tool. Ultimately, BERT achieves (previously) unrivaled linguistic context understanding [140]. However, as an encoder-only model with immense contextual knowledge, BERT is not trained to do specific subsequent tasks, such as sentiment analysis, named entity recognition, text summarization, or machine translation [136]. Practitioners must *fine-tune* a base BERT model: fine-tuning, a common practice in language modeling, involves additional training atop a base model on a specific task [136]. Figure 4 (**Pre-train—Fine-tune**) visualizes this workflow. Fine-tuning requires an additional model layer for the task, a fine-tuning dataset (e.g., a question and answer dataset for Q&A fine-tuning), and brief training of the new layer.

BERT's pre-training provides vast knowledge of linguistic context, making task-specific fine-tuning relatively accessible. As an early and influential PLM, BERT has many off-the-shelf variants[8] available. Lightweight variants, such as DistilBERT [141] and ALBERT [142], have helped lower model inference[9] costs. Other variants, such as RoBERTa [143], modify the pre-training objective to improve performance. The wide availability of BERT (and variants) allows practitioners with limited compute resources to perform otherwise inaccessible tasks.[10] However, while BERT is small by LLM standards, data collection, cleaning, and model training are still very expensive.

Alongside BERT, other model families were developed, such as OpenAI's decoder-only GPT (Generative Pre-trained Transformer) models, which helped explore the impact of increasing model size. The (2018) GPT model [117] featured 117M parameters (comparable to BERT and also pre-trained on BookCorpus) and the (2019) GPT-2 model [126] with variants up to 1.5B parameters. This sets the stage for the infamous GPT-3 model (Sec. 3.2) [116]. As we will see, the decoder-only GPT models perform better and better in conversational settings in which the user provides information in context, as opposed to the encoder-only BERT relying upon itself for context.

[5] See https://mkofinas.github.io/tikz_tutorial.pdf for the visualization template.

[6] BERTBASE and BERTLARGE have 110M and 340M parameters, respectively. Note: we will often abbreviate M = Million, B = Billion, T = Trillion when discussing large numbers.

[7] The acronyms LMs (language models) and PLMs (pre-trained language models) are often used alongside "LLMs"

[8] See https://huggingface.co/models?other=bert for a staggering number of variants.

[9] Here we distinguish model inference - using the model to perform a task - from model training.

[10] This underscores the cost of training truly large (by current standards) language models far exceeding BERT's size.

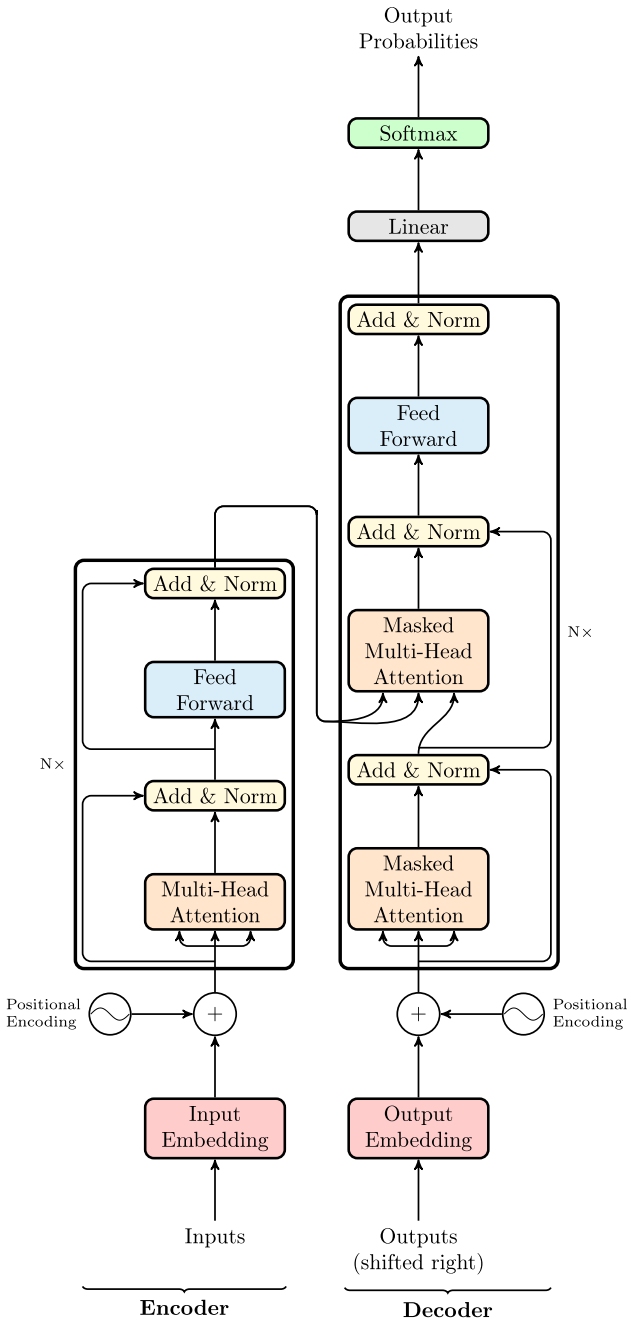


Fig. 3 The Transformer Architecture. The transformer encoder (left) and decoder (right). Transformers come in three variants: encoder-decoder, encoder-only, and decoder-only

3.2 Large language models

While smaller models, such as BERT, are effective on many NLP tasks, increasing parameters to produce so-called large language models (LLMs) has proven to be a fruitful area of study [128]. LLMs are versatile, performing many NLP tasks often near-SOTA (state-of-the-art). Similar to BERT, large models are trained on massive text data and can be applied to many downstream tasks (often through fine-tuning) [137]. Unlike BERT, however, many of these models are not easily run on consumer hardware. Given the model sizes, resource-rich organizations tend to build large models with less resource-rich organizations (or resource-conscious large organizations) fine-tuning large models, training smaller models, or co-opting large models to improve smaller models.

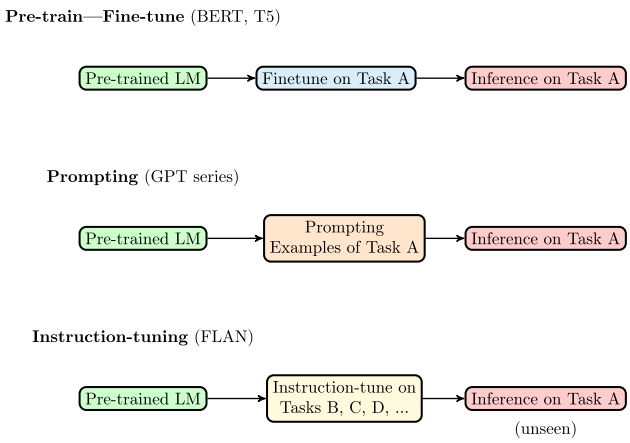


Fig. 4 Inference paradigms for pre-trained LMs [135]. Pre-train—Fine-tune consists of fine-tuning a PLM (e.g., BERT, T5) on task (A) and performing inference on task (A). Prompting consists of using a PLM (e.g., GPT models) without fine-tuning and performing inference on task (A). Instruction-tuning consists of instruction-tuning a PLM (e.g., FLAN) on many tasks (B, C, D,...) and using the model for inference on a *different* task (A)

Though a demarcation for *large* LMs is fuzzy, we begin by further discussing GPT-2 [126], briefly mentioned in Sec. 3.1. As a decoder-only model, GPT-2[11] can be applied to NLP tasks without fine-tuning, yet variants up to 1.5B parameters achieved at/near-SOTA zero-shot performance [126]. Zero-shot refers to the number of examples (zero) provided to the model, explaining the task at hand: models are often evaluated in zero-shot, one-shot, and few-shot settings. For example, a user may provide English to French translations to "explain" the desired task before giving an English excerpt to translate — this is *in-context learning*. Figure 4 (Prompting) visualizes this workflow.

Given strong zero-shot performance, [126] highlights the impact of large, high-quality training data, not just increased model size. This allows a reframing of specific tasks, such as translation or Q&A, which would otherwise require fine-tuning (e.g., with BERT), into generic language modeling. To illustrate, English to French translation can be modeled "as the sequence (translate to french, english text, french text)", where the model predicts "french text" [126]. A similar framing of, for example, Q&A associated with a given passage can be used. The authors argue that a high quality dataset features "naturally occurring demonstrations" of such tasks, representing an important shift in perspective on the need for fine-tuning foundation models. The strong general performance of GPT-2, at its relatively large size, motivated further exploration into model scaling.

An important exploration into both model scaling and data usage is Google's T5 (Text-to-Text Transfer Transformer) model[12] [144]. T5 is an (up to) 11B parameter encoder-decoder model, larger than BERT and GPT models discussed so far, that frames language tasks as "text-to-text" modeling by structuring data as input/output text pairs. This can be done by "embedding" the task into the data; for example, Q&A data might be structured as (input, output) = (question: Q, answer: A). Similar to GPT models, T5 can perform many NLP tasks without task-specific fine-tuning. Furthermore, [144] introduces the C4 (Colossal Clean Crawled Corpus) dataset, a cleaned version of the Common Crawl,[13] which has become a prominent dataset in language modeling [145].

The major achievement of further scaling is showcased in *Language Models are Few-Shot Learners*, where GPT-3, a 175B parameter successor[14] to GPT-2, demonstrates strong performance gains, especially with in-context learning, while successfully performing NLP tasks on which it was not explicitly trained [116]. Structurally similar to GPT-2, several GPT-3 variants of increasing size (up to 175B parameters) were trained to examine the impact of model size. The GPT-3 variants were examined with a provided prompt and examples of the desired task (e.g., English to German translations) during dialog without any fine-tuning. The use of examples helped "coach" the model into completing the prompt and, for some tasks, performance improved with more examples provided. The larger (175B) GPT-3 was able to achieve at or near-SOTA performance on several tasks/benchmarks. Importantly, GPT-3 was not fine-tuned on these tasks while other SOTA methods were, underscoring the generality of GPT-3. GPT-3's successor, GPT-4 (March 2023) has been widely hailed for its SOTA performance and abilities [146] available via API [147], though model details (e.g., size, training data) are

[11] See the model release at https://openai.com/research/gpt-2-1-5b-release.

[12] T5 is available at https://github.com/google-research/text-to-text-transfer-transformer.

[13] https://commoncrawl.org/

[14] The GPT-3 paper was released on arXiv circa May-July 2020 (https://arxiv.org/abs/2005.14165) and later at NeurIPS 2020 [116]. The model was not released.

scant. GPT-4 has since been extended to allow visual input with GPT-4V(ision)[15]) [148] (Sept 2023) and audio and visual with GPT-4o[16] (May 2024), helping drive multi-modal (e.g., text, vision, and image) models into the mainstream. Recently, OpenAI has produced the GPT-o1[17] model and previewed the GPT-o3 model.

In keeping with model secrecy, GPT-4 is rumored to use a Mixture-of-Experts (MoE) approach, a modeling style in which multiple "experts" are trained and a query is routed to the appropriate expert [149]. This approach has gained traction, including use in xAI's open source (quite large—314*B* parameters) Grok-1[18] (March 2024) and more restricted Grok-2[19] (August 2024) models. Another major development towards model generality is the FLAN (Finetuned Language Net) approach in which PLMs (up to 137B parameters) undergo instruction-tuning (i.e., fine-tuning for instructions) [135]. Essentially, task-oriented NLP datasets featuring natural language instructions are used for fine-tuning: fine-tuning is done on all tasks except A, then evaluation is done on A. The models learn to follow instructions on both seen and *unseen* tasks. Figure 4 (**Instruction-tuning**) visualizes this workflow. Remarkably, zero-shot FLAN outperforms few-shot GPT-3 in many settings and few-shot FLAN outperforms zero-shot. Interestingly, instruction-tuning is effective only for large models (68B+ parameters) and negatively impacts smaller models. Scaling FLAN to larger models and datasets is shown to be effective in [150].

While the GPT series was incredibly successful, intuitively interacting with LLMs remains difficult for non-technical users as models are (typically) not *chatbots* — they are not necessarily designed to be conversational. This limitation is inherently problematic for the layperson and greatly impedes widespread adoption. The eminently popular ChatGPT[20] alleviates many of these conversational pain points. Beyond mere conversational ability, alignment with human values is important. As such, OpenAI has investigated *reinforcement learning from human feedback* (RLHF) following GPT-3 to produce a new model, InstructGPT, aligned with human motives and values [151]. Other efforts at alignment, such as Anthropic's use of *Constitutional AI* [152, 153] to produce the Claude[21] (March 2023), Claude 2[22] (July 2023), and Claude 3[23] (March 2024) models have also been successful.

Although OpenAI's GPT family represents a major thread of development, LLM development has been a community-wide effort. Google, for instance, has trained the LaMDA[24] (Language Models for Dialog Applications — Jan. 2022) family of models featuring up to 137B parameters [154]. On the larger side, Google's PaLM (Pathways Language Model — Apr. 2022), a massive 540B parameter model, achieved SOTA performance on many benchmarks in the few-shot setting [155]. More recently, in keeping with GPT-4's secrecy, Google has produced the Gemini (Dec. 2023) [156] and Gemini 1.5 (Feb. 2024) [157] models, with unspecified sizes such as "Pro", "Flash", and "Ultra." These decoder-only models have progressed similarly to the GPT family and have served as the backend of Google's Bard chatbot[25] [158].

Major advancements by OpenAI and Google have not been open source (often available via API access), making reliable evaluation difficult [159]. However, meaningful open source development has taken place. Similar in size to GPT-3, the decoder-only BLOOM (BigScience Large Open-science Open-access Multilingual) language model is something of an outlier in terms of development, representing a massive coordinated effort across many participants and organizations [160]. Additionally, the OPT (Open Pre-trained Transformer) language models are a "suite of decoder-only pre-trained transformers ranging from 125M to 175B parameters" [161]. These models tend to be structurally inspired by GPT-3 and the larger models (i.e., 176B and 175B parameters) perform similarly to GPT-3 in many cases, but the data secrecy and exact architecture of GPT-3 prevent precise examination into divergences. Other meaningful contributions explore the search space of model size, data, and compute, all key aspects of model performance. Given a (training) compute budget, the tradeoff between data size and model size is examined in [129].

[15] https://openai.com/contributions/gpt-4v

[16] https://openai.com/index/hello-gpt-4o/

[17] https://openai.com/index/introducing-openai-o1-preview/

[18] https://x.ai/blog/grok-os

[19] https://x.ai/blog/grok-2

[20] Available at https://openai.com/chatgpt with version ChatGPT−3.5 in the non-subscription plan at time of writing (https://platform.openai.com/docs/model-index-for-researchers). It should be noted that the exact composition of the underlying ChatGPT LLM is not disclosed, although the lessons of the GPT models are clearly in play.

[21] https://www.anthropic.com/index/introducing-claude

[22] https://www.anthropic.com/index/claude-2

[23] https://www.anthropic.com/news/claude-3-family

[24] https://blog.google/technology/ai/lamda/

[25] https://blog.google/technology/ai/bard-google-ai-search-updates/

Meta released the (not fully) open-source LLaMA (Large Language Model Meta AI) [131] and LLaMa 2 [162] models[26] in February and July of 2023, respectively. Smaller than GPT-3 at the top end, the LLaMA models are of size 7, 13, 33, and 65 billion parameters. Large models such as GPT-3 tend to perform well, but the shear size makes inference costs, ignored in [129], prohibitive for the typical end user. Thus, as the LLaMA models would be released to researchers (i.e., researchers must perform inference), the Meta team decided to train comparatively smaller models for longer. Later, Meta released the open source LLaMa 3 "herd" of models,[27] beginning with LLaMa 3[28] (April 2024), LLaMa 3.1[29] (July 2024), and LLaMa 3.2[30] (Sept. 2024) [163].

The smaller LLaMA models have proven quite popular as targets of fine-tuning, for example, LLaMA 7B is fine-tuned for code review in [164] and medical dialog in [165]. While LLaMa 1 & 2 are permissively licensed, they are not fully open source; thus, open source variants (OpenLLaMA-7B and OpenLLaMA-13B) have been produced [132]. A notable example of fine-tuning LLaMA *and* co-opting a larger model is the development of the Alpaca model [166]. LLaMA 7B and 13B are fine-tuned with instruction examples obtained by an OpenAI GPT-3+/InstructGPT variant in a cost effective manner. Thus, not only is the base model (LLaMA) pre-trained, but the fine-tuning exploits a large, capable model. Alpaca subsequently shows major performance improvements over its LLaMA base [166]. Another fine-tuned LLaMA model, Vicuna-13B, was trained on "user-shared ChatGPT conversations" and demonstrated strong conversational abilities [167].

Just as smaller models have been fine-tuned by less resource-rich organizations, larger models have been similarly adapted. For example, Google has extended the PaLM models into the medical domain with Med-PaLM [168] and Med-PaLM 2 [169], providing powerful tools for assisting highly qualified professionals[31] [170]. Other high risk applications, such as law, are steadily coming into the LLM purview as GPT-4 can pass the Bar Exam [171] and perform well on the GMAT (Graduate Management Admission Test) [172].

Given the capabilities and real (or at least perceived) usefulness of these models across diverse applications, concerns about their use have arisen. For instance, social and economic concerns, especially around labor market impact, have been notable [173]. However, a major impediment to widespread adoption is the *hallucination* problem, whereby a model produces factually incorrect content during routine interactions, posing problems for misinformation and high risk applications [174–176]. Especially considering the breakneck speed of recent advances, broader concerns of AI safety, especially artificial general intelligence (AGI), have come to the forefront of research [177]. As LLMs have become pervasive conversational companions, humanistic concerns, such as the psychology [178, 179], personality [180], and bias [181] of models are also being explored.

While the knowledge encoded into LLMs is useful for many downstream tasks, the availability of up-to-date information remains a challenge [182]. Recall, these models are trained on large text corpora, but do not undergo continual "re-training." Some tasks, such as writing advertising copy, are not sensitive to outside information. However, sufficiently dated models might provide stale/inaccurate information or be fundamentally unhelpful for rapidly evolving domains (e.g., news). Naturally, there are tasks that fall between these extremes, such as summarizing an article (which can be provided to the model in-context). This constraint foreshadows a potential struggle of applying LLMs for RS, as the systems don't necessarily have current knowledge.

Finally, with this detailed background given, we additionally provide Fig. 5 as a visual guide of language models over the past several years.[32] The figure broadly demonstrates the increasing diversity and specialization of language models, with major AI research labs such as OpenAI, Google DeepMind, Meta, Anthropic, Mistral, and Baidu driving innovation. The models are categorized into three major branches: encoder-only, decoder-only, and encoder-decoder, as previously discussed in Fig. 3.

The encoder-only branch (left, orange) covers model discussed in Sec. 3.1, such as BERT [125], DistilBERT [141], ALBERT [142], and RoBERTa [143], as well as a few additional models for reference, such as ELECTRA [184] and the recent ModernBERT [185]. Similarly sparse, the little-used complete encoder-decoder architecture is given by the right, yellow branch,

[26] See https://ai.meta.com/blog/large-language-model-llama-meta-ai/ and https://www.llama.com/llama-downloads/ for descriptions and model access.

[27] With sizes from 1B, 3B, 9B, 11B, 70B, 90B, and 405B parameters.

[28] https://ai.meta.com/blog/meta-llama-3/

[29] https://ai.meta.com/blog/meta-llama-3-1/

[30] https://www.llama.com/

[31] https://sites.research.google/med-palm/

[32] See [183] for the inspiration of this figure.

which traces back to Meta's BART [186] and Google's T5 [144], evolving into Flan-T5 [187], mT5 [188], Ernie 4.5[33], and Ernie X1.[34]

The decoder-only branch (middle, green) illustrates many of the models discussed in this section, such as GPT [117], GPT-2 [126], GPT-3 [116], GPT-4 [147], as well as recent iterations, such as GPT-4o [189] and GPT−4.5.[35] Other significant models we've discussed are also shown, such as PaLM [150], BLOOM [160], OPT [161], the LLaMA series [131, 162, 163], Alpaca [166], Vicuna [167], the Claude series[36] (including the recent Claude 3.7 Sonnet[37]), various Gemini models [156, 157] (including the recent Gemini 2.0 Flash[38]), the Grok series,[39] as well as a handful of important models we did not directly cover, such as Mixtral 8x7B [190], the DeepSeek series [191, 192], Manus [193], Hunyuan [194], Doubao,[40] Qwen [195], and QwQ.[41]

In the next section, we discuss *prompting*, general techniques for interacting with conversational LLMs, which allows users to navigate conversations and provide in-context information to models.

3.3 Model prompting

The input and instruction provided to a conversational model, or *prompt*, are integral to eliciting a quality response [196]. As LLMs are difficult and expensive to train, much research has been focused on prompting, which is easier and more cost-effective to manipulate. Perhaps the most famous example of successful prompting is in-context learning with GPT-3 [116]. With modern chatbots, such as ChatGPT, Bard, and Claude trained to engage conversationally and follow instructions, in-context learning is often not required.[42] Nevertheless, the lesson of providing clear, descriptive instructions still applies. Done successfully, prompting is a convenient way to coax an LLM into performing specialized tasks. For example, a user can tell an LLM to assume the identity of a subject matter expert, thereby improving its responses related to the subject [197] or even to critique its own responses, allowing the system to iteratively improve upon its own output [198].

Even so, models frequently fail to capture intermediate reasoning steps, particularly in multi-step problems, resulting in so-called "one-step missing errors" [199]. Chain-of-Thought (CoT) prompting was developed to encourage LLMs to structure multiple logical steps to reach conclusions. Originally, this meant providing solved examples of problems in the prompt, with the goal of enticing the LLM to use a similar logic structure when solving new problems [199]. Further, CoT-structured training data is shown to be very effective for instruction-tuning with the PaLM model in [150]. Self-consistent CoT improves CoT by sampling several reasoning paths and examining the logical consistency of the paths [200]. The recently proposed Tree of Thoughts system [201] aims to further improve CoT prompting by modeling each step in the chain as a separate thought and conducting a tree search over possible thoughts, with the LLM itself critiquing the thoughts in order to prune the tree. While these strategies cover only a glimpse of prompting research, advanced strategies often involve inserting structure into the conversation and giving the LLM "lots of compute / many tokens" to produce an answer, as demonstrated above, as opposed to forcing the model to produce an answer quickly. This appears to give models "room to reason" and thereby provide better quality responses.

While the aforementioned techniques help to extract information from LLMs, it is often the case that an LLM is incapable of providing a quality answer by itself. For example, an LLM trained with data up to 2022 would likely provide poor information on current geopolitical affairs.[43] Even with relatively frequent model updates, LLMs will struggle on rapidly evolving domains of inquiry, such as news. Of course, these pain points can be partially mitigated via prompting by, for example, providing a current document/article to the model and asking about its content.

[33] https://www.prnewswire.com/news-releases/baidu-unveils-ernie-4-5-and-reasoning-model-ernie-x1–makes-ernie-bot-free-ahead-of-schedule-302402490.html

[34] It is unclear whether the Ernie series models are encoder-decoder based or of any other architecture. Thus they are shown with dotted lines in the figure.

[35] See https://openai.com/index/introducing-gpt-4-5

[36] https://www.anthropic.com/index/introducing-claude

[37] https://www.anthropic.com/news/claude-3-7-sonnet

[38] See https://blog.google/technology/google-deepmind/google-gemini-ai-update-december-2024/

[39] https://x.ai/news/grok-os, https://x.ai/news/grok-2, https://x.ai/news/grok-3

[40] https://team.doubao.com/en/publication

[41] https://qwenlm.github.io/blog/qwq-32b-preview/

[42] Non-chat models should be handled individually, respecting the model's documentation to ensure a successful interaction.

[43] Ideally, the system will inform users of such limitations.

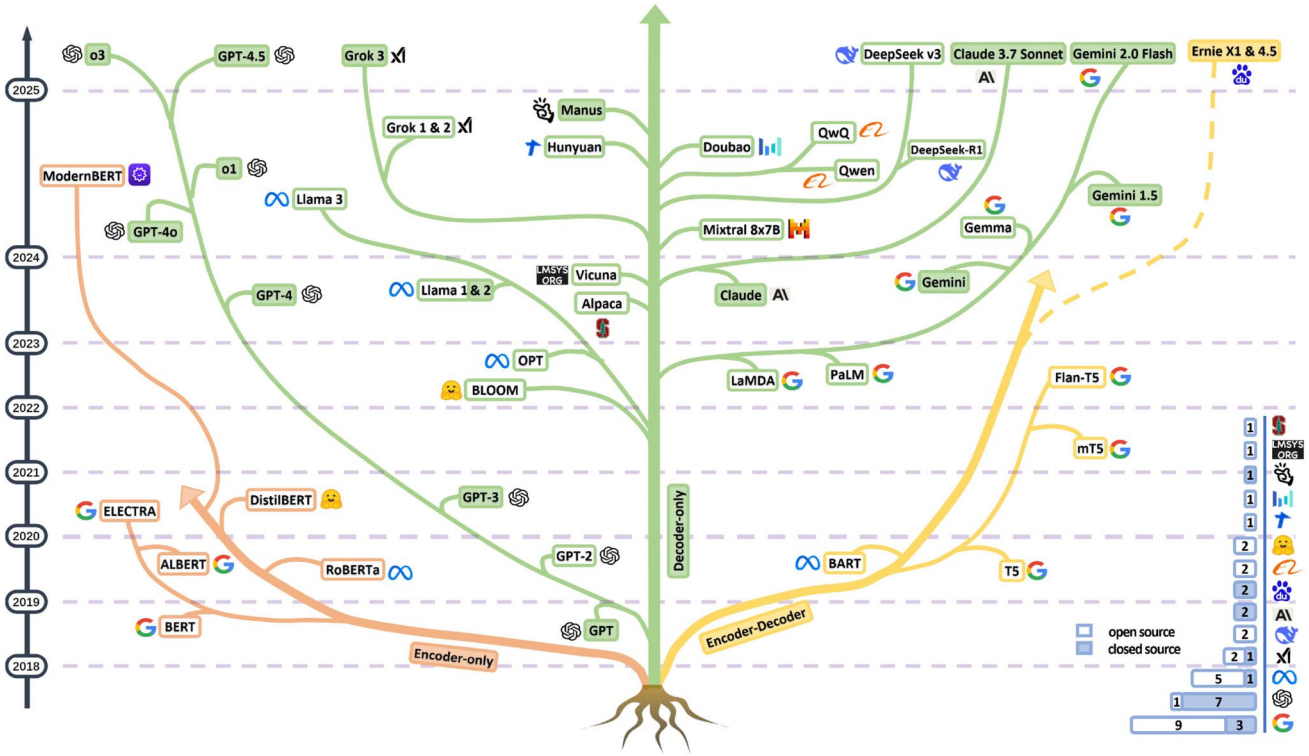


Fig. 5 Evolution of Language Models in the recent decade. The tree diagram illustrates the development of major language models, categorized into Encoder-only (orange, left-branch), Decoder-only (middle-branch, gree), and Encoder-Decoder (right-branch, yellow) models. The figure highlights the progression from early foundation models such BERT and GPT to large modern releases including Gemini 2.0, GPT–4.5, and DeepSeek. We include a logo beside each model to the developer. The Y-axis indicates the year of release of the models. The hollow boxes are open-source models, while the solid boxes are closed-source models. The stacked bar plot in the bottom right corner shows the number of major models released by various developers

Retrieval-Augmented Generation (RAG), first proposed in 2020, aims to improve a model's generative abilities by providing access to external data [202]. This generally includes a collection of documents containing knowledge (e.g., Wikipedia) and a pre-trained retriever attached to a standard LLM. The retriever encodes a user's query and pulls relevant documents. The documents are combined with the user's original query to form a single prompt. This allows the LLM to access a large base of information without retraining. Additionally, document citations can be provided. Naturally, this helps to factually ground model results even when the concern of outdated information is not present. New research in this area improves on the RAG framework by training an LLM to know when to ask for more information [203]. Multiple possible documents are then added to the prompt to generate a range of possible outputs, which the LLM critiques to select the best. Similar in spirit to RAG, LLM's have become a target of integration with search engines, where the model essentially searches the internet and synthesizes a response to user queries [204].

3.4 Practical challenges and concerns

Although LLMs are incredibly versatile and powerful, they inherit a number of limitations. From the standpoint of building and using such models, computational costs are extreme [205]. Large models typically have tens, or even hundreds, of billions of parameters. These models require specialized and expensive GPU clusters to train and perform inference at scale. This lack of scalability is being addressed in two primary ways. First, as discussed in Sec. 3.1, (much smaller) encoder-only models continue to show promise in specialized tasks via fine-tuning. Second, smaller decoder-only models (especially in edge environments) are increasingly capable, though they often require fine-tuning [206].

With models built and available, perhaps the chief advantage of LLMs—the diversity and flexibility of the output—leads to many enormous challenges [207]. The models are a reflection of their training data, which can lead to inadvertent discovery and reinforcement of problematic patterns. Bias [208–210] and fairness concerns [211, 212], including

unequal access [213], persist, particularly on the basis of race, gender, and age. Such biases can lead models to produce stereotypes and other harmful information.

Perhaps the most publicly known type of potentially harmful information is hallucinations, where a model produces factually incorrect content [214–217]. While hallucinations are sometimes amusing,[44] they can pose risks in high-stakes environments such as medicine [218] where there is little room for error, but also in everyday decision making [219].

Beyond generating factually incorrect information, models are notoriously difficult to update with new knowledge. "Catastrophic forgetting" describes the fragility of an LLM's knowledge when exposed to new data [220, 221]. A lack of continual learning can lead to outdated and incorrect content over time. Even for correct content that is not stale, there is still a lack of transparency in decision-making and explainability [222, 223]. In the opposite direction of updating a model with new information is machine *unlearning*, the process of removing knowledge from a model,[45] is often difficult [224].

An interesting wrinkle in producing factually incorrect information can be found in the use of LLMs to solve math problems. The models are historically quite bad at simple math problems [225], but have shown promise recently [226]. Poor performance can be attributed to a number of reasons, such as tokenization and limited mathematical training data. However, model flexibility shines here, as models can be used to in a CoT-esque approach to use programs and tools to break down problems and solve them systematically [227]. This highlights an important aspect of LLMs: they are not good at everything, but can learn to use tools to compensate [228].

Security and privacy further complicate the use and deployment of LLMs [215, 229–231]. Adversarial prompting [232, 233] and jailbreaking [215, 229] are two approaches used to exploit LLMs by uncovering shortcomings and circumventing safeguards and constraints, respectively.[46] Such attacks can be severe, with consequences "ranging from disinformation to data breaches and reputation damage" [229]. Correctly noted in [234], the closed-source and centralized nature of many foundation models poses a threat and barrier to adoption where "many mission-critical and copy-righted data cannot allow centralized data collection and training." As highlighted by [231], security and privacy threats impact diverse domains, such as transportation, education, and healthcare.

While aforementioned limitations of LLMs are relatively apparent, a recent modeling development has framed the autoregressive nature of these models as a weakness [235]. Diffusion LLMs seek to reframe language modeling as a diffusion problem in which response text is masked (recall, masking as described in BERT's training) and incrementally unmasked. In this sense, a model's response is considers the whole context at once, as opposed to content generated sequentially.

4 Language models and recommendation

Sec. 3 provides a broad, yet brief, overview of recent language modeling advances. That exposition serves as the foundation for a discussion of the intersection of language models and the recommendation task. Given the flexibility and interactivity of LLMs, it is natural to apply these models to the RS domain [236, 237]. Just as we discussed models of varying sizes, including small and large models, and architectures, including encoder-decoder, encoder-only, and decoder-only models, we discuss these models in the recommendation context. Transformers can be used to directly power sequential recommendation and transformer-based LLMs are applied to recommendation in two primary ways: as end-to-end recommenders and as encoders for recommendation. Figure 6 provides a visual for these approaches.

4.1 The transformer architecture for sequential recommendation

Recommender systems have long adopted neural network architectures to facilitate recommendation [238]. Occasionally, standard architectures such as recurrent neural networks perform well compared to older, non-neural methods, but the outperformance is hardly conclusive [239, 240]. Recent research has adopted more modern architectures, including

[44] For example, a recommendation to put glue on pizza given here: https://www.theverge.com/2024/5/23/24162896/google-ai-overview-hallucinations-glue-in-pizza

[45] For example, this would be required in the "right to be forgotten" in the European Union's General Data Protection Regulation (GDPR) regulations (https://gdpr.eu/right-to-be-forgotten/.

[46] Red-teaming is the practice of adversarial prompting and jailbreaking done ethically, with the intent of bolstering the model's guardrails and further understanding limitations [215].

the transformer [122], as seen in Fig. 6. While the following approaches are not technically language models as they are not trained on natural language, we discuss them as they are fundamentally linked to LLM structure and development.

The sequential, token-based nature of modern LLM architectures makes them well suited for sequential recommendation tasks. In the language modeling task, a model is given a sequence of tokens (i.e., words or word pieces) and asked to predict the next (or missing) token. Likewise, in the sequential recommendation task, a model is given a series of items a user has interacted with and asked to predict the next item.

The first major transformer-based recommendation model was SASRec [241], released in 2018. SASRec is *causal*, meaning it is trained only on items before the item it is being asked to predict. It achieved marked performance improvements of up to 25% on NDCG@10 over MF and earlier neural models and has been widely adopted as a benchmark in the field. Later, BERT4REC [242] expanded on this success. Taking inspiration from BERT (Sec. 3.1) [125], BERT4REC is *bidirectional* – trained on both sides of an item token, rather than just the items coming before it. This appears to help the model understand overall user behavior better, as BERT4REC often outperforms SASRec, with up to a 15% improvement on NDCG@10.

More recent work has focused on incremental improvements on these architectures. Some have augmented the base SASRec architecture with additional features, including temporal awareness [243] and user personalization [244]. The attention mechanism at the core of the transformer architecture has also been a subject of research, with several papers [245, 246] modifying it to better suit the recommendation task by adding cross attention, which trains attention on the interactions between users and items. This has resulted in notable performance gains over self-attention only transformers. The important S3-Rec model [247] improved transformer-based models such as SASRec and BERT4Rec by adding self-supervision and found notable performance gains, including improvements of between 6% and 25% on NDCG@10 over the nearest competitor across various media domains. Other research has focused on elucidating the reasons for BERT4REC's apparent performance improvements over SASRec [248, 249] and on improving the architecture when data are noisy [250].

The datasets used to train and test LLM RS vary widely. Much work, of course, is done in-house at large media corporations using proprietary data. However, several major public datasets are frequently used. Perhaps most important is MovieLens, an expansive collection of datasets containing movie rankings, ratings, and user data [251]. The Amazon Product Reviews dataset is also very popular, especially given its wide variety of product categories [252]. For news recommendation, Microsoft's MIND dataset is widely cited [253]. Finally, the Last.FM dataset is a public collection of music listening data that is often used to test implicit feedback for RS [254].

As new advancements in LLM architecture are developed, sequential recommendation is likely to advance in parallel. Table 2 summarizes the main LLM-based RS cited here.[47]

4.2 LLMs as end-to-end recommenders

One very prominent approach to using LLMs for recommendation has been to treat LLMs as a complete end-to-end recommendation system, as seen in Fig. 6. Put simply, this entails asking an LLM for a recommendation, ranking, or rating, and sampling its output in natural language. This approach has become popular in light of the major advances in "foundation models" such as GPT-3, as discussed in Sect. 3.2. These models encode immense knowledge in their parameters and have been shown to be effective at many diverse problems beyond language modeling [137]. They are suited to the tricky task of conversational recommendation in natural language. However, unlike earlier models such as BERT and GPT-2, many cutting-edge foundation models are not open source, and even open source models can be very expensive to run or fine-tune when large. Therefore, it can be difficult to obtain embeddings or perform recommendation fine-tuning. Using foundation models as black boxes, end-to-end recommenders avoid this problem.

Using LLMs as end-to-end recommenders began in earnest with the 2019 release of GPT-2 [126]. While GPT-2 was open-source and therefore did not have to be used in an end-to-end fashion, it was the first generative LLM large enough to be viable for this task. Earlier attempts had been made to co-opt the encoder-only BERT model's ability to predict the order of sentence pairs for conversational recommendation but found that it was mostly incapable of performing true end-to-end conversational recommendation when faced with previously unseen sentences, even after fine-tuning [269]. GPT-2, on the other hand, showed more ability [255–257]. This early work gave GPT-2 an input sequence of items and asked it to provide similar items as output. GPT-2 fulfilled this task relatively well, mostly giving relevant and accurate

[47] Not included are models such as SASRec and BERT4REC which apply the Transformer architecture but do not use pretrained language models.

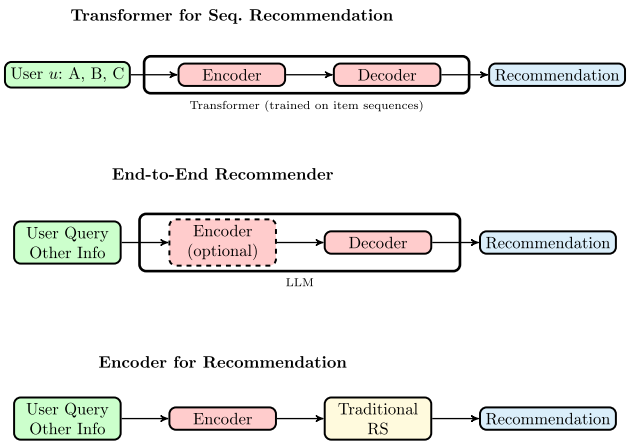


Fig. 6 Three primary ways in which the Transformer is used for recommendation. The Transformer for Sequential Recommendation approach involves training a transformer on item consumption sequences. A user *u* will receive a recommendation based on their item sequence A, B, C. The End-to-End Recommender approach involves taking a user query and side information as input into an encoder-decoder or decoder-only (dashed line indicates optional Encoder) model that produces recommendations. The Encoder for Recommendation approach involves putting a user query and side information into a pre-trained encoder and using the output encodings as input into a traditional RS algorithm that produces recommendations

recommendations. However, major problems with the end-to-end approach were uncovered: it is extraordinarily difficult to disentangle the recommendation task from the LLM's primary purpose of providing realistic, natural dialogue. This leads to several issues. LLMs will not consistently provide output in any structured format, and sometimes they will give non-recommendation output in an attempt to continue the conversation. They also have a strong bias towards more grammatical output over recommendation, sometimes to the point of "hallucinating" movies, books, or songs that don't exist. Both base and fine-tuned GPT-2 still mostly underperformed classical MF methods [256].

Research on LLMs as end-to-end recommenders has dramatically accelerated in the past year following the release of OpenAI's ChatGPT[48] chatbot, which garnered widespread public attention. Suddenly, ChatGPT and similar chatbots have made end-to-end LLM recommendation easily and cheaply/freely accessible to users. In 2023, several studies began to evaluate the recommendation capabilities of ChatGPT [103, 258, 259]. They found that ChatGPT[49] clearly outperform previous LLMs at recommendation. In settings with very sparse training data or user cold starts, ChatGPT outperforms older techniques such as MF and more modern approaches such as neural collaborative filtering [103]. However, ChatGPT is still outperformed by even simple algorithms like k-nearest neighbors in more data rich contexts [259]. ChatGPT's immense knowledge base and natural language understanding enable it to achieve cutting-edge recommendation performance as an end-to-end recommender when recommendation-specific data is lacking, but its lack of specificity means it is outperformed by algorithms more tailored to recommendation when more data is available.

As research on end-to-end recommendation with LLMs has accelerated, it has also diversified. Early papers attempted only to quantify the overall performance of foundation models, often quickly covering the separate tasks of item recommendation, rating prediction, and item ranking together and evaluating LLMs in comparison to classical systems. Recent research has moved from broad comparisons with classical systems to specific areas of interest. Fine-tuning has been a major line of investigation, with the TALLRec framework proposed to efficiently fine-tune LLMs for recommendation [260]. The T5 model [144] has been a popular choice for fine-tuning which, when fine-tuned for recommendation, can perform competitively with classical approaches [237, 261]. Kang et al. [270] focused on the task of rating prediction and evaluated several major LLMs, finding that even very large models like PaLM underperform MF and other modern CF approaches. However, they find a T5 model fine-tuned for rating prediction can outperform even very modern collaborative filtering based approaches. Ho et al. [262] investigated LLMs as item rankers, finding that large models consistently produce correctly formatted output and rank items competitively when the lists of candidate items are small. The recent RecBench benchmark compared both off-the-shelf and fine-tuned LLMs to traditional recommender systems. It found that after fine-tuning, LLMs recommendation rankings can achieve improvements of up to 170% in the NDCG@10 metric [271]. A

[48] Available at https://openai.com/chatgpt.

[49] ChatGPT is an interface for interacting with an underlying LLM, which changes over time. ChatGPT currently uses version of GPT-4 for the free version, with access to more advanced variants, including the o1 and o3 models in the paid version.

benchmarking focused specifically on recent versions of ChatGPT, including ChatGPT-4, found that even without fine-tuning it could quantitatively outperform more traditional RS. Compared to the highest-performing traditional approach in each category, ChatGPT showed improvements of 95% in nDCG and 83% in recall for the books domain. However, ChatGPT performed worse in the music and movie domains, possibly relating to its advantage in understanding text compared to visual and audio data [259]

On the prompting side, the Chat-REC system was able to improve ChatGPT's recommendation performance by constructing standardized prompts incorporating user queries, user profiles and histories, and the output of more traditional recommender systems into a single LLM input [263]. The idea of using traditional RS to produce candidate items for LLM-based recommendations has garnered interest, including proposals to apply the popular retrieval-augmented generation framework [202] to recommendation [264]. In the world of evaluation, Zhang et al. [272] studied the problem of fairness in LLM recommendation. It is known that LLMs "inherit stereotypes, misrepresentations, derogatory and exclusionary language, and other denigrating behaviors that disproportionately affect already-vulnerable and marginalized communities" [273]. This is also true in the recommendation context, where ChatGPT is found to vary its recommendations when it knows users are members of sensitive groups, resulting in potential unfairness. The problem of fairness in recommendation is already well known; including LLMs and their innate bias could increase the potential for bias further. ChatGPT is further analyzed as an "Artificial General Recommender" (AGR) in [274], mirroring the famous "Sparks of AGI" research [146]. The AGR is a theoretical RS capable of advanced recommendation on any topic in many forms of interaction while also obtaining information from users and being resilient to mistakes and inconsistencies. The authors evaluate ChatGPT as a prospective AGR. They find the system has deficiencies, such as bias towards more recent input information and not addressing user inconsistencies/contradictions, but the system generally performs well given the state of the field. Given the recent surge in interest in LLMs, it seems likely cutting-edge LLMs will continue to be important in the field of recommendation.

4.3 LLMs as encoders for recommendation

While LLMs have proved remarkably capable as end-to-end RS, there remain large problems with that approach. LLMs can be hard to fine-tune to a specific recommendation task, have a limited range of input mechanisms, often produce output in inconsistent formats, and struggle to take advantage of recommendation-specific context data. Solutions to these issues are often awkward. For example, retrieval-augmented generation [202, 264] relies on using a different system to find candidate outputs and then pasting that list of candidates into the LLM's input. Given the non-integrated nature of these kinds of solutions, and the strong performance of other, more traditional RS, there has been substantial interest in more elegantly integrating traditional RS and LLMs to take advantage of the strength of both techniques. This usually means using only the encoder side of the LLM, utilizing the powerful capability of LLMs to translate text into numerical representations (embeddings) but leaving the final recommendation task up to more traditional RS. This broad structure is illustrated in Fig. 6.

Interest in using LLM embeddings for recommendation began as an attempt to solve the item ID problem in recommendation. Often, collaborative filtering models learn based on item IDs, arbitrarily assigned numerical codes that uniquely identify items (think "Blue Tennis Shoes: Product Code 44243"). These IDs are usually assigned within each item dataset, and there is usually no carryover between datasets: a movie, book, or product might be assigned one ID in one dataset but another ID in a different dataset. This makes building universal models very difficult and means that collaborative filtering models must be trained from scratch on each new dataset. The early ZESRec model [265] attempted to solve this problem by using BERT to obtain universal embeddings for each item. Rather than working based off of pre-assigned item IDs, ZESRec uses BERT embeddings as the IDs, as seen in Fig. 7. These embeddings are plugged into a traditional neural network-based sequential recommendation model. The authors found ZESRec to outperform even neural methods trained on the target data, demonstrating the power of universal embeddings in capturing information about the world. Later models such as UniSRec [266], VQ-Rec [267] and Recformer [275] further developed this approach; Recformer was able to improve NDCG@10 performance by a further 9% to 30% over earlier methods like SASRec and S^3-Rec [275]. While both the encoders and RS models vary, all follow a similar approach: use an LLM encoder to generate embeddings from item names and descriptions, then train a traditional ID-based RS on those embeddings. One downside of this approach is that the resulting embeddings are essentially meaningless outside of the context of the model that created them – there is currently no known way to assign any kind of qualitative meaning to a single element of the embedding vector. Like all LLMs, the embedding model is essentially a "black box." Despite the limitation, survey work

has confirmed the embedding approach to be competitive with ID-based RS [276], and finds that scaling and fine-tuning the encoder model used can further improve performance [277].

Recognizing the power of LLM embeddings, recent work has expanded their use beyond just ID. The advanced CTRL model [278], itself inspired by Amazon's early CTR-BERT [279], integrates outputs from both LLM and collaborative filtering models into input for a single traditional model which is then fine-tuned. This also allows for context information to be input into the LLM. The KAR model [268], employed by Huawei, specifically focuses on also incorporating LLM reasoning and context knowledge into more traditional RS; the idea of using LLM reasoning in RS is further explored in [280]. Multiple LLMs can also be combined: [281] fuses GPT-derived embeddings into pretrained models like BERT, while UniTRec [282] predicts user-item scores through a transformer-based encoder and decoder. Given the increasing ability of end-to-end LLM recommendation models, there has even been interest in integrating end-to-end models with collaborative filtering [283] or even adding embeddings obtained from traditional models to LLMs, rather than vice versa [284]. And Rajput et al. even find that transformers can be trained to generatively predict these embedding IDs [285]. The synergistic power of LLMs and traditional RS is likely only to grow as both fields advance.

4.4 Miscellaneous uses of LLMs for recommendation

In addition to these core uses, there have been many other proposals for incorporating LLMs into recommendation. One emerging area of research is data generation. Scarcity of data is often a problem for RS training, especially conversational RS. LLMs have previously been used to generate data in other contexts; recent research has begun to expand this to the field of RS. Synthetic data is often used in training [286, 287] but also in generating prompts based on user information for other LLM-based RS [268]. Using LLMs for data generation also allows user privacy to be preserved, given that real-world training data often contains sensitive information [288].

LLMs can also be adopted to better model users. Their large size and deep understanding of the world allow for a more holistic picture of users. [102] focuses specifically on capturing changes in user interests over time using LLMs, while [289] uses LLMs to generate synthetic user profiles based on review text. And multiple roles can be combined: [290] focuses on integrating many different roles for LLMs into one large, unified RS. With LLMs currently leading the charge towards more intelligent computer systems, their capabilities will likely be leveraged for RS in more and more creative ways.

5 Conclusions

The intersection of RS and LLMs is a burgeoning field of research with ongoing innovation and progress. To appreciate the breadth of opportunities, we began our discussion with classical RS, including approaches such as collaborative filtering, content-based methods, and knowledge-based systems, which have long served as the *de facto* methods for online personalization. Although other approaches, such as conversational recommender systems, have long appeared promising, the challenges from the relative lack of conversational recommendation data to the complex natural language requirement had largely inhibited successful implementation [111]. This can now be partially mitigated by employing LLMs to produce realistic synthetic training data.

Our discussion of classical recommender systems focused on the types of data used for recommendations. For collaborative filtering, communal data (often in ratings matrix form) is used. For content-based methods, item information (without communal context) is used, and for knowledge-based systems, knowledge data (about products) is used. While knowledge-based systems sometimes have a level of interactivity, these classical approaches largely produce predictions surfaced to the user (e.g., via panels on streaming platforms). Conversational recommender systems differ in *how* the recommendations are made — conversationally. That is, classical approaches usually operate on tabular data and output point predictions, while modern conversational systems must process natural language inputs and outputs, requiring coherent and relevant responses. Such systems are useful in many applications, since natural language is inherently very flexible. Regardless of the mode of interaction, modern recommender systems are an integral part of the contemporary online experience, helping users sift through enormous volumes of content.

Given the broad industry success enjoyed by RS, emerging models are quickly adopted for use. Centrally important to our exploration is the transformer architecture and subsequent research involving RS. Research has advanced very quickly, both in breadth and depth. Underpinning cutting-edge language models and performing exceptionally well in many diverse natural language tasks, the transformer has been applied far and wide for recommendation.

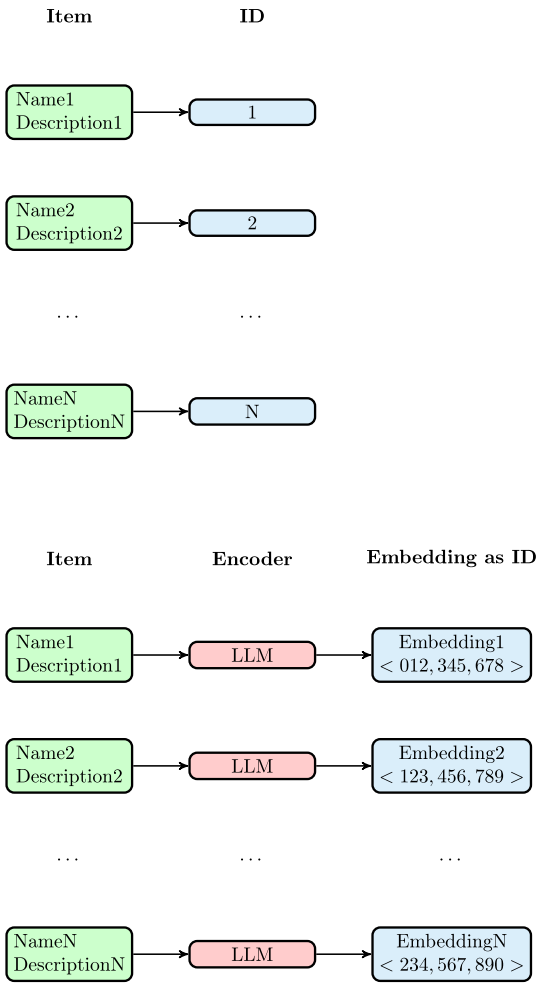


Fig. 7 In traditional RS models, items are assigned unique IDs arbitrarily, often sequentially. These IDs contain no information, serving only as unique references to the underlying item. In encoder-based RS, an LLM translates some text related to the item, such as name and website description, into an embedding (a unique vector capturing the text's semantic meaning). These embeddings not only uniquely identify the item but also contain information about it, endowing an otherwise entirely collaborative approach with contextual information

Of particular note, pre-trained language models, especially decoder-only chatbots, are a promising approach to solving recommendation problems. Well-versed in language and proficient across NLP tasks, the potential for circumventing conversational recommendation data shortage is especially appealing [135]. The pre-training of these models represents an interesting transitional trend of classical inflexible models to general-purpose models and is echoed in the generality of the data used for modeling. We see classical CF and CB approaches using well-defined, structured data during training, KB approaches using reference data, and finally, LLMs using general web-scale information with possible fine-tuning or access to external data.

Even while LLMs are powerful conversational agents, they do not strongly outperform classical RS approaches in general. However, they may provide a desirable level of flexibility, allowing the user to express their current interests in their own words [270], which may in the future permit an enriched recommendation, although challenges remain. Consider, for example, the use of LLMs for media recommendation, in particular content on streaming services. While someone may visit a chatbot (e.g., ChatGPT) for a film recommendation, knowing where to then consume the film (i.e., what platform) is an additional problem. Streaming platform catalogs are often not public, so it is unlikely ChatGPT will have accurate, up-to-date information (for non-original content). Direct integration of LLMs into conventional RS applications (e.g., Netflix, Hulu) is thus a possible step into the future of RS: users would simply describe the content they wish to consume and the platform can provide relevant recommendations. Undoubtedly, both public and private RS research, powered by open- and closed-source LLMs will eventually handle many real-world recommendation tasks, increasing their end-user utility.

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Table 2 Summary of Major LLM-based Recommender Systems and Their Attributes

System	Approach	Underlying LLM/Model	Key Attributes/Features	Reference
GPT-2 as RS	End-to-End	GPT-2	Uses the off-the-shelf GPT-2 model as RS with a custom prompting framework	[255–257]
ChatGPT as RS	End-to-End	ChatGPT (GPT−3.5/4)	Uses off-the-shelf ChatGPT as RS with a custom prompting framework	[103, 258, 259]
TALLRec	Fine-tuned End-to-end	Llama 7B	Applicable to many LLMs	[260]
T5 Recommender	Fine-tuned End-to-End	T5	Fine-tunes earlier T5 LLM for RS	[237, 261]
LLM Zero-shot Ranker	End-to-end ranking	GPT−3.5 Turbo	Competitively ranks candidate items in RS context	[262]
Chat-REC	End-to-End	GPT−3.5 Turbo	Uses standardized prompts that combine user queries, profiles, and outputs from a traditional RS	[263]
RAG-based LLM RS	End-to-End with RAG	GPT−3.5-turbo	Grounds recommendations by retrieving external documents to augment the prompt	[264]
ZESRec	Embedding-based	BERT	Uses BERT embeddings to encode item features for a CB system. Important early work	[265]
UniSRec	Embedding-based	BERT	A more universally applicable embedding CB approach	[266]
VQ-Rec	Embedding-based	BERT	Improves on the embedding paradigm by adding an intermediate vector "code" layer between item text and embedding	[267]
KAR	End-to-End	Various	A framework for incorporating 1) LLM reasoning about users and 2) LLM factual knowledge about items into any RS	[268]

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Declarations

Ethics approval and consent to participate Not applicable.

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